

Status Finished

Started Thursday, 30 October 2025, 12:45 PM

Completed Thursday, 30 October 2025, 1:10 PM

Duration 25 mins 25 secs

Marks 1.89/3.00

Grade 1.89 out of 3.00 (62.98%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Incorrect

Mark 0.00 out of 1.00

Find the solution to the initial value problem

$$\begin{cases} a_n + 4 \cdot a_{n-1} + 3 \cdot a_{n-2} = 0 \\ a_0 = 5 \\ a_1 = -13 \end{cases}$$

$a_n =$

Input your answer as a function of n . Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **$2^n + 3$** .

Your last answer was interpreted as follows:

$$(-1)^n + 4 \cdot (-3)$$

The variables found in your answer were: $[n]$

Question 2

Partially correct

Mark 0.89 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle 1, 2, 3 in a counter-clockwise manner. The symmetry $r1$ is rotation by 120 degrees, sending vertex 1 to vertex 2. The symmetry $r2$ is rotation by 240 degrees. The element $t1$ is the reflection symmetry that fixes vertex 1, and likewise for $t2$ and $t3$.

The multiplication table for this group looks like

id	r1	r2	t1	t2	t3
r1	r2	id	.	.	.
r2	id	r1	.	.	.
t1	t3	t2	id	r2	r1
t2	t1	t3	.	.	.
t3	t2	t1	r2	r1	id

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3 .

Syntax help: Remember that we compose symmetries from left to right. So if a and b are two symmetries, then the composite $a \cdot b$ is the symmetry that first applies a , and then b .

$t2 \cdot t1 =$	r1
$r2 \cdot t1 =$	t3
$r1 \cdot t1 =$	t2
$t2 \cdot t2 =$	id
$r2 \cdot t2 =$	t1
$r1 \cdot t2 =$	t3
$t2 \cdot t3 =$	r1
$r2 \cdot t3 =$	t2
$r1 \cdot t3 =$	t1

Your last answer was interpreted as follows:

r1

Your last answer was interpreted as follows:

t3

Your last answer was interpreted as follows:

t2

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t1

Your last answer was interpreted as follows:

t3

Your last answer was interpreted as follows:

r1

Your last answer was interpreted as follows:

t2

Your last answer was interpreted as follows:

t1

Question **3**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing products and inverses in the group $(P_{45}, \bullet_{45})$ of integers and multiplication modulo 45.

$$44 \bullet_{45} 29 = \boxed{16}$$

Your last answer was interpreted as follows:

16

$$29 \bullet_{45} 28 = \boxed{2}$$

Your last answer was interpreted as follows:

2

$$28 \bullet_{45} 37 = \boxed{1}$$

Your last answer was interpreted as follows:

1

$$44^{-1} = \boxed{44}$$

Your last answer was interpreted as follows:

44

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 508

Signing code:

Your last answer was interpreted as follows:

21132

